Cloud Computing

Chapter 5 – Cloud Billing

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Introduction

- Cloud computing helps businesses store data and run applications online
- It offers scalability and flexibility
- However, **cost management is critical**
- Cloud billing helps monitor and control expenses

What is Cloud Billing?

Definition

Cloud billing is the procedure of producing and sending invoices for using cloud services to customers. Most cloud provider carriers provide special billing reports, permitting groups to apprehend their intake patterns. These reports provide data on the resources used, the time period of utilization, and associated expenses. By comprehensively studying these reports, companies can benefit from insights into their expenditure and make informed decisions to optimize their utilization and reduce costs.

- Process of generating invoices for cloud service usage
- Provides detailed reports of:
  - Resource usage
  - Time duration
  - Associated costs
- Helps organizations analyze and optimize spending

Key Benefits of Cloud Billing

- **Cost Transparency:** Provides detailed insights into usage, helping organizations understand where their money is spent.
- **Cost Control:** Identifies unused or underutilized resources, enabling reduction of unnecessary expenses.
- **Forecasting and Budgeting:** Uses historical data to predict future costs and plan budgets effectively.
- **Resource Optimization:** Analyzes usage patterns to ensure efficient utilization of cloud resources.
- **Cost Allocation:** Distributes costs across departments for better financial management and accountability.

Steps to Monitor Costs (GCP)

1. Access GCP Console
2. Go to Billing Section
3. Set Up Billing Account
4. Link Projects
5. Access Billing Reports

Advanced Monitoring Steps

6. Set Budgets and Alerts
7. Export Data to BigQuery
8. Optimize Resources
9. Monitor and Fine-Tune
10. Collaborate and Review

What is Google Cloud Platform (GCP)?

- **Google Cloud Platform (GCP)** is a cloud computing service provided by Google.
- It allows businesses and developers to:
  - Run applications
  - Store and manage data
  - Handle workloads efficiently
- Built on **secure, scalable, and high-performance infrastructure**
- Supports technologies like:
  - Virtual Machines
  - Kubernetes (container management)
  - BigQuery (data analytics)
  - TensorFlow (machine learning)
- Helps organizations:
  - Reduce costs
  - Improve performance
  - Scale without managing physical servers

Google Cloud vs Google Cloud Platform (GCP)

Aspect	Google Cloud	GCP
Definition	Broad suite of services by Google for digital transformation	Subset of Google Cloud providing public cloud infrastructure
Key Services	Includes Workspace, Android Enterprise, Chrome OS, ML APIs	Compute, storage, networking, databases
Example Offerings	Gmail, Docs, Drive, Android, APIs	Compute Engine, App Engine, Cloud Storage, Kubernetes, BigQuery
Target Use Case	Collaboration tools, device management, enterprise APIs	Hosting and scaling applications in the cloud

Billing Reports

- Cost Overview
- Cost Details
- Custom Reports

Purpose:

- Understand spending patterns
- Identify cost-heavy services

Google Cloud Services Overview

Category	Service	Description
Cloud Computing	Compute Engine	Create and manage virtual machines with control over resources like memory, storage, and security.
Cloud Computing	Google Kubernetes Engine (GKE)	Managed Kubernetes service for deploying apps with automatic scaling and load balancing.
Cloud Computing	App Engine	Scalable platform for running web applications with automatic demand handling.
Storage	Cloud Storage	Stores large amounts of data with high availability and easy access.
Storage	Persistent Disk	Durable storage attached to virtual machines for flexible usage.
Storage	Cloud SQL	Fully managed database service supporting MySQL, PostgreSQL, and SQL Server.
Networking	Virtual Private Cloud (VPC)	Provides a private network environment for secure application deployment.
Networking	Cloud Load Balancing	Distributes traffic across instances for better performance.
Networking	Cloud CDN	Delivers cached content from nearest location for faster access.
Data Analytics	BigQuery	Data warehouse for fast analysis of large datasets.
Data Analytics	Dataflow	Processes and analyzes data pipelines efficiently.
Data Analytics	Pub/Sub	Messaging service enabling asynchronous communication between systems.
Machine Learning	Vertex AI Platform	Platform for building and managing AI models.
Machine Learning	AI Platform Training	Enables training of machine learning models in the cloud.
Machine Learning	AI Platform Prediction	Provides predictions using trained ML models.
Productivity	Google Workspace	Tools like Gmail, Drive, and Calendar for collaboration.
Security	Cloud IAM	Controls access to resources ensuring security and authorization.

Cloud Cost Optimization & Best Practices

Definition: Process of reducing cloud costs by monitoring usage and applying strategies like **rightsizing, auto-scaling, and reserved instances**

Need for Cloud Cost Optimization:

- **Pay-per-use Pricing:** Pay only for resources used
- **Dynamic Resources:** Avoid over/under provisioning
- **Avoiding Waste:** Eliminate unused resources
- **Cost Management:** Control increasing cloud expenses
- **Value Maximization:** Get maximum ROI from cloud
- **Cost Transparency:** Clear visibility of spending
- **Scalability:** Efficient scaling without overspending
- **Resource Allocation:** Focus on critical workloads
- **Vendor Management:** Optimize multi-cloud costs

Best Practices to Reduce Cloud Bills

1. **Identify Mismanaged Resources**
   - Detect over/under-utilized resources
   - Remove unused resources
   - Optimize storage usage
2. **Reserved Instances**
   - Long-term commitment (1–3 years)
   - Lower cost than on-demand
3. **Auto-Scaling**
   - Add/remove resources automatically
   - Scale as per demand
4. **Right-Sizing**
   - Match resources with workload
5. **Real-Time Analytics**
   - Monitor usage instantly
   - Make quick decisions
6. **Cost Anomaly Detection**
   - Identify cost spikes
   - Fix root causes
7. **Automate Right-Sizing**
   - Optimize during provisioning
8. **Remove Unused Resources**
   - Delete snapshots, stop idle services
9. **Evaluate Providers**
   - Compare pricing and performance

Numerical 1

Scenario: A cloud service guarantees **99.9% uptime**. Calculate the **maximum allowed downtime per year**.

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Solution:

- Total hours/year $= 365 \times 24 = 8760$
- Downtime $= 0.1\%$ of 8760

$$= 0.001 \times 8760 = 8.76 \text{ hours}$$

Answer: 8.76 hours

Numerical 2

Scenario: A cloud provider charges $Rs.10/\text{hour}$. A system runs unnecessarily for 50 extra hours.

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Solution:

$$\text{Extra Cost} = 50 \times 10 = Rs.500$$

Answer: $Rs.500$ loss due to unused resources

Numerical 3

Scenario: A VM costs $Rs.20$/hour (on-demand) and $Rs.12$/hour (reserved). Calculate savings for 100 hours usage.

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Solution:

$$\text{On-demand} = 100 \times 20 = Rs.2000$$

$$\text{Reserved} = 100 \times 12 = Rs.1200$$

$$\text{Savings} = Rs.800$$

Answer: $Rs.800$ saved

Numerical 4

Scenario: A system auto-scales from 10 to 4 instances during off-peak hours. Each instance costs $Rs.5/\text{hour}$. Duration $= 6$ hours.

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Solution:

$$\text{Without scaling} = 10 \times 5 \times 6 = Rs.300$$

$$\text{With scaling} = 4 \times 5 \times 6 = Rs.120$$

$$\text{Savings} = Rs.180$$

Answer: $Rs.180$ saved using auto-scaling

Case Study Numerical 1

Scenario: A company uses cloud services with **99.9% SLA uptime**. Total usage = **1 month (30 days)**. Actual downtime recorded = **5 hours**. Cost of service = **Rs.1000/hour**. Penalty clause: **10% refund if SLA violated**. **Find:**

- Allowed downtime
- Whether SLA is violated
- Total cost and refund

Case Study Numerical 1

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- Whether SLA is violated
- Total cost and refund

Solution:

- Total hours $= 30 \times 24 = 720$
- Allowed downtime $= 0.1\% \times 720 = 0.72$ hours
- Actual downtime $= 5$ hours > 0.72 hours
- $\Rightarrow$ SLA Violated
- Total cost $= 720 \times 1000 = Rs.7{,}20{,}000$
- Refund $= 10\%$ of $7{,}20{,}000 = Rs.72{,}000$

Answer: SLA violated, refund $= Rs.72{,}000$

Case Study Numerical 2

Scenario: A streaming company runs 20 VMs at $Rs.15/\text{hour}$. Using auto-scaling, VMs reduce to 8 for 10 hours daily. Month = 30 days.
Find:

- Cost without auto-scaling
- Cost with auto-scaling
- Total savings

Case Study Numerical 2

Scenario: A streaming company runs 20 VMs at $Rs.15$/hour. Using auto-scaling, VMs reduce to 8 for 10 hours daily. Month = 30 days.

Find:

- Cost without auto-scaling
- Cost with auto-scaling
- Total savings

Solution: Without Auto-Scaling:

$$20 \times 15 \times 24 \times 30 = Rs.2,16,000$$

With Auto-Scaling:

- Peak (14 hrs): $20 \times 15 \times 14 \times 30 = Rs.1,26,000$
- Off-peak (10 hrs): $8 \times 15 \times 10 \times 30 = Rs.36,000$

$$\text{Total Cost} = 1,26,000 + 36,000 = Rs.1,62,000$$

$$\text{Savings} = 2,16,000 - 1,62,000 = Rs.54,000$$

Answer: Savings = $Rs.$54,000

Case Study Numerical 3

Scenario: A cloud provider guarantees **99.95% uptime**. A company uses the service for **1 year**. Actual downtime recorded = **6 hours**. Penalty: **5% refund if SLA violated**. Total annual bill = **Rs.10,00,000**.

Find:

- Allowed downtime
- SLA violation status
- Refund amount

Case Study Numerical 3

Scenario: A cloud provider guarantees **99.95% uptime**. A company uses the service for **1 year**. Actual downtime recorded = **6 hours**. Penalty: **5% refund if SLA violated**. Total annual bill = **Rs.10,00,000**.

Find:

- Allowed downtime
- SLA violation status
- Refund amount

Solution:

- Total hours/year = $365 \times 24 = 8760$
- Allowed downtime = $0.05\% \times 8760$

$$= 0.0005 \times 8760 = 4.38 \text{ hours}$$

- Actual downtime = 6 hours > 4.38 hours
- $\Rightarrow$ SLA Violated
- Refund = 5% of $Rs.10,00,000 = Rs.50,000$

Answer: SLA violated, refund = $Rs.50,000$

Case Study Numerical 4

Scenario: A company runs 15 instances at $Rs.12$/hour. Due to poor monitoring, 5 instances remain idle for 8 hours daily. Month = 30 days. **Find:**

- Total wasted cost due to idle resources
- Optimized cost after removing idle instances

Case Study Numerical 4

Scenario: A company runs 15 instances at $Rs.12$/hour. Due to poor monitoring, 5 instances remain idle for 8 hours daily. Month = 30 days. **Find:**

- Total wasted cost due to idle resources
- Optimized cost after removing idle instances

Solution:
Wasted Cost:

$$5 \times 12 \times 8 \times 30 = Rs.14,400$$

Total Cost Without Optimization:

$$15 \times 12 \times 24 \times 30 = Rs.1,29,600$$

Optimized Cost:

$$(10 \times 12 \times 24 \times 30) = Rs.86,400$$

Savings:

$$1,29,600 - 86,400 = Rs.43,200$$

Answer: Wasted cost = $Rs.$14,400, Savings = $Rs.$43,200

Budgets and Alerts

- Set spending limits for projects
- Get alerts when thresholds are crossed
- Helps prevent overspending

Resource Optimization

- Identify unused resources
- Resize instances (rightsizing)
- Use auto-scaling
- Reduce unnecessary costs

Tips for Effective Cloud Billing

- Regularly review billing reports
- Use budget alerts
- Automate resource scaling
- Collaborate with finance and IT teams

Best Practices

- Set clear budgets
- Use resource tagging
- Monitor usage patterns
- Conduct regular review meetings
- Use cost management tools

Common Issues

- Unexpected cost spikes
- Unused resources
- Poor cost allocation
- High data transfer costs

Troubleshooting

- Analyze billing reports for spikes
- Tag resources properly
- Remove unused services
- Optimize data transfer